Simulation Arena

Deal Aging & Multi-Quarter Sales Cycles: The Pipeline Hopper

Sales says quarter 2; the odds from your history are 0.00041% for closing then - 0.00061% of the value slips past that date, and >99.9% never closes at all. Plan on $32 of this $3.00M deal, not the full amount. That is after allowing for this deal having sat 5 quarters - a deal that had just arrived would be at 0.0029%.

Seed 42 | Run run_f3f2eb4e8854 | 100,958 scenarios tested | Markov chain on your own stage-to-stage… | 22 Sep 2026, 07:44

Same seed, same numbers - this run can be reproduced exactly.

How this works

Every deal on your board hops from stage to stage. We counted how your own deals actually hopped - forward, staying put for a quarter, or dying - and turned those counts into odds. We also account for how long a deal has been sitting still: the older it gets, the more likely it is to slip or die, so a deal that has been parked for a year is priced differently from one that has just arrived. Then we rolled the odds forward one quarter at a time, which gives the chance of the deal being anywhere on the board in each future quarter. Money is then weighted by those odds: if there is a 1-in-8 chance of closing next quarter, only an eighth of the value deserves to be in next quarter's plan.

The numbers that matter

CHANCE IT CLOSES IN Q2, AS SALES PROMISED
0.00041%
A slim chance the money actually lands in the quarter it was promised for.
vs if we ignored how long it has been sitting: 0.0029%

CHANCE IT SLIPS PAST THAT QUARTER
0.00061%
The chance the deal does close, but later than the date sales promised.

CHANCE IT DIES INSTEAD
>99.9%
The deal never closes - it stalls or the customer walks.
vs 0.0023% still stuck in Legal at the end of the horizon

WHAT YOU CAN BANK ON BY Q2
$32
The deal value weighted by its real chance of arriving - 0.0% of $3.00M.

CASH COMMITTED AGAINST ODDS
$3.00M
You are planning to spend $3.00M that has only a 0.0% chance of arriving.
vs $3.00M promised vs $32 realistic

QUARTERS UNTIL IT REALLY CLOSES
3.5
On average, counted from today, for the deals that do close (3.2 quarters to a final yes or no).
vs sales promised 2

HOW LONG THIS DEAL HAS BEEN SITTING
5 quarters
The older a deal gets, the more likely it is to slip or die - we have priced that in.
vs we treat it as stale after 4 quarters

How we got there

1. What we measured

We read 100,958 real stage moves from your history and turned them into odds: from each stage a deal moves forward, sits still for a quarter, or dies.

Evidence: 100,958 observed moves

2. The forecast versus the odds

Sales has this deal closing in Q2. The odds from your history put it at 0.0% for landing in that quarter - and a >99.9% chance of never landing at all.

Evidence: 0.0% vs 100%

3. Where the time actually goes

Deals do not walk through the stages, they sit in them. Looking across the whole horizon, 0.0% of the deals that close do so later than Q2, and 0.0% close after this fiscal year has ended. Even at the end of Q6, 0.0% are still sitting in Legal.

Evidence: 0.0% stuck in Legal

4. Why the age of the deal matters

This deal has been running 5 quarters, and we priced it that way. A deal that had only just arrived would show 0.0% for Q2 instead of 0.0%. Deals that sit still are the ones that quietly die, which is why age is part of the forecast rather than an afterthought.

Evidence: 0.0% at 5 quarters vs 0.0% when fresh

5. The number to plan with

Weighted by the odds, this deal is worth $32 to Q2's plan, not $3.00M. The difference, $3.00M, is money you would be committing before it has any real chance of arriving.

Evidence: $3.00M of over-commitment

6. What to do about it

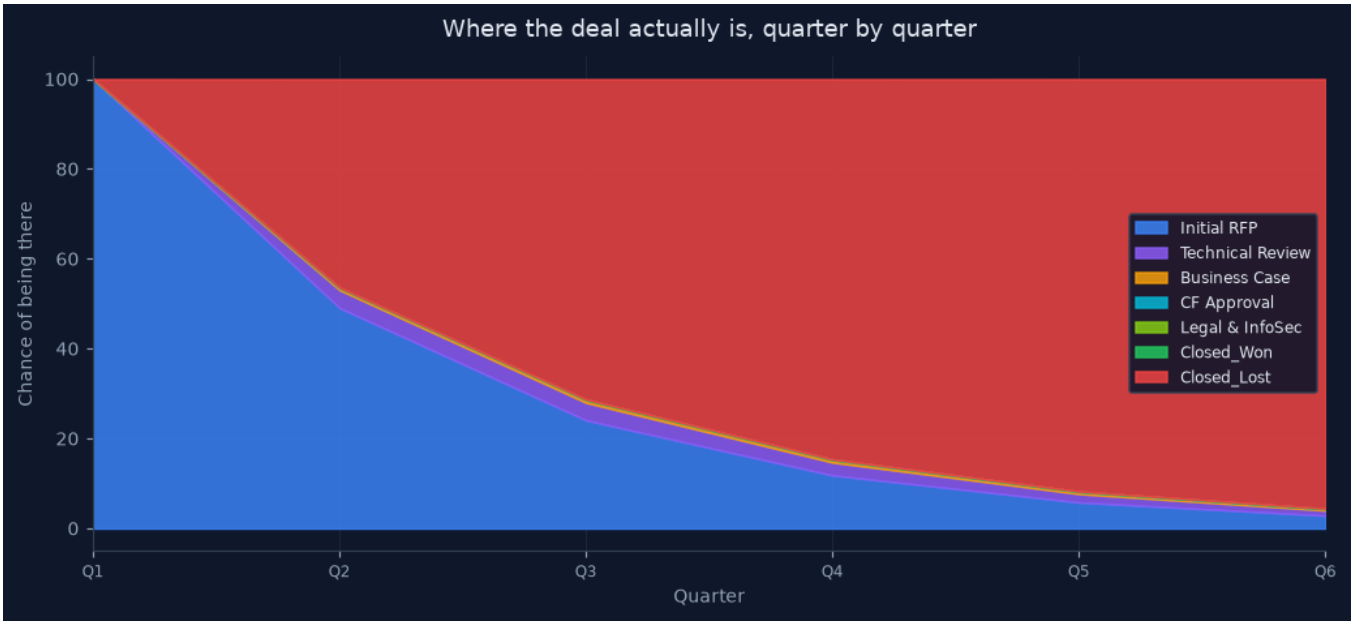
Plan the cash against $32, hold the rest as a contingency, and put the deals that have been sitting the longest in front of whoever owns the stage they are stuck in. A forecast that says 'Q2' with no odds attached is not a forecast - it is a hope.

Evidence: plan on 0.0% of the value

Charts

Where the deal actually is, quarter by quarter

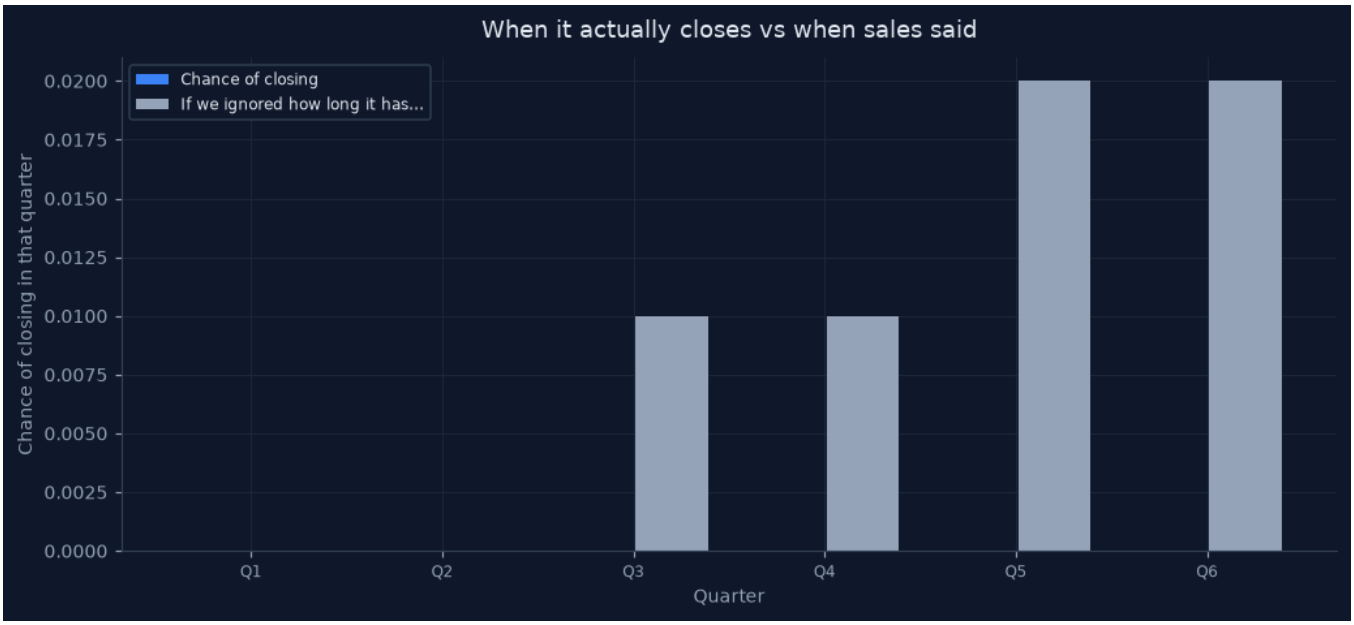
Starting from 'Initial RFP'. Green is won, red is dead.



What this shows: Read the green band: that is the chance the deal is already closed by each quarter. The grey blocks are the stages it is still sitting in.

When it actually closes vs when sales said

Chance of closing in each quarter; sales promised Q2



What this shows: The tall bar is the quarter the deal is most likely to close in. It is almost never the quarter on the forecast.

Money you can actually plan on

Cumulative deal value weighted by its chance of arriving



What this shows: The gap between the green line and the red line is cash you would be committing before it has arrived.

Tables

Quarter by quarter: where the deal is

The chance of the deal sitting in each stage at the end of every quarter.

Quarter	Initial RFP	Technical Review	Business Case	CF Approval	Legal & InfoSec	Closed_Won	Closed_Lost
Q1	49.1%	4.0%	0.48%	0.055%	0.00066%	0.00066%	46.4%
Q2	24.1%	3.9%	0.63%	0.093%	0.003%	0.0011%	71.3%
Q3	11.8%	2.9%	0.57%	0.1%	0.004%	0.0013%	84.6%
Q4	5.8%	1.9%	0.45%	0.091%	0.0038%	0.0015%	91.8%
Q5	2.8%	1.1%	0.32%	0.072%	0.0031%	0.0016%	95.6%
Q6	1.4%	0.67%	0.21%	0.053%	0.0023%	0.0016%	97.7%

Read the 'Closed_Won' column to see how the chance of closing builds up.

The money, weighted by the odds

What this deal is really worth to each quarter's plan.

Quarter	Chance of closing then	Value earned that quarter	Value banked by then	Chance closed by then	Chance dead by then	Still not banked
Q1	0.00066%	$20	$20	0.00066%	46.4%	$3.00M
Q2	0.00041%	$12	$32	0.0011%	71.3%	$3.00M
Q3	0.00025%	$8	$40	0.0013%	84.6%	$3.00M
Q4	0.00015%	$5	$44	0.0015%	91.8%	$3.00M
Q5	<0.0001%	$3	$47	0.0016%	95.6%	$3.00M
Q6	<0.0001%	$2	$48	0.0016%	97.7%	$3.00M

Row 2 is the quarter sales promised.

What sitting still costs you

The same deal judged at three ages, plus the age we actually used for it.

How long it has been sitting	Chance it closes by Q2	Chance it never closes
Fresh (under 2 quarters)	0.025%	98.1%
Settled (2-4 quarters)	0.01%	>99.9%
Stale (over 4 quarters)	<0.0001%	100.0%
This deal (5 quarters)	0.0011%	>99.9%

The older a deal is, the lower it sits. The highlighted row is the age we used for the forecast above.

Your open deals, weighted the same way (top 25)

Every open deal on the board, priced by the odds of it actually closing - each one judged at its own age.

Deal	Stage now	Value	Closes next quarter	Within 2 quarters	Ever closes	Expected quarter	Value you can bank on	Quarters sitting
P-00011589	Business Case	$40.00M	0.08%	0.29%	3.5%	6.7	$1.40M	1
P-00038587	Initial RFP	$40.00M	0.0071%	0.025%	1.9%	9.5	$762k	0
P-00054552	Business Case	$40.00M	<0.0001%	<0.0001%	<0.0001%	n/a	$0	7
P-00047903	Initial RFP	$40.00M	0.0071%	0.025%	1.9%	9.5	$762k	0
P-00054879	Technical Review	$40.00M	0.0042%	0.053%	2.3%	8.3	$932k	1
P-00042281	Technical Review	$36.62M	0.0042%	0.053%	2.3%	8.3	$853k	1
P-00027130	Business Case	$35.98M	0.08%	0.29%	3.5%	6.7	$1.26M	1
P-00051237	Initial RFP	$32.70M	0.0052%	0.01%	0.042%	5.2	$14k	2
P-00068394	Initial RFP	$32.42M	0.0071%	0.025%	1.9%	9.5	$618k	0
P-00081129	Technical Review	$29.82M	0.0083%	0.017%	0.062%	4.5	$19k	3
P-00032218	Initial RFP	$29.58M	0.0071%	0.025%	1.9%	9.5	$564k	1
P-00004141	Initial RFP	$28.96M	0.0052%	0.01%	0.042%	5.2	$12k	3
P-00026774	CF Approval	$28.48M	<0.0001%	<0.0001%	<0.0001%	n/a	$0	9
P-00049379	Technical Review	$27.33M	0.0083%	0.017%	0.062%	4.5	$17k	2
P-00059817	Initial RFP	$26.98M	<0.0001%	<0.0001%	<0.0001%	n/a	$0	11
P-00083844	Initial RFP	$26.24M	0.0052%	0.01%	0.042%	5.2	$11k	2
P-00065601	Technical Review	$25.72M	0.0083%	0.017%	0.062%	4.5	$16k	2

Deal	Stage now	Value	Closes next quarter	Within 2 quarters	Ever closes	Expected quarter	Value you can bank on	Quarters sitting
P-00072655	Business Case	$25.04M	0.08%	0.29%	3.5%	6.7	$874k	1
P-00095862	Initial RFP	$24.60M	0.0071%	0.025%	1.9%	9.5	$469k	0
P-00085009	Technical Review	$24.57M	0.0042%	0.053%	2.3%	8.3	$572k	1
P-00075948	Technical Review	$24.35M	0.0042%	0.053%	2.3%	8.3	$567k	0
P-00004222	Legal & InfoSec	$24.34M	0.85%	1.2%	1.5%	1.7	$362k	3
P-00044677	Business Case	$24.14M	0.08%	0.29%	3.5%	6.7	$842k	1
P-00072186	Technical Review	$24.07M	0.0083%	0.017%	0.062%	4.5	$15k	2
P-00080629	Initial RFP	$24.05M	0.0071%	0.025%	1.9%	9.5	$458k	0

Value you can bank on = deal value x its chance of ever closing.

What we recommend

Do not book $3.00M for Q2. Book $32 and hold the rest back.

Risk High 100 out of 100 | Confidence: High

Why

• The chance of closing in Q2 is 0.0%, not 100%.

• 0.0% of the value arrives later than promised. and >99.9% never arrives at all.

• Planning on the full $3.00M commits $3.00M of cash against a 0.0% chance.

• The deal has been running 5 quarters; at that age a deal is judged 0.0% likely to close by Q2 if it were fresh, so sitting still is costing real odds.

What to do

1. Put $32 in the Q2 plan and treat the remaining $3.00M as contingent on the deal actually reaching the next stage.

2. Attach a stage gate to any hiring, inventory or spend that depends on this deal.

3. Re-forecast the whole open pipeline the same way, not just this deal - the same optimism is almost certainly in the rest of the board.

4. Ask sales to forecast with odds attached, not dates.

The trade-off: Booking less now means a smaller number in this quarter's plan - but it avoids the mid-year freeze that follows a promise that does not arrive.

What we assumed and what to watch

Worth knowing before you act

• Sales says Q2; the model says 0.0%. Committing the full $3.00M on that date is the single biggest risk in this plan.

• >99.9% of deals at this stage never close at all - they are not 'slipping', they are gone. Treat that part of the pipeline as zero.

• This deal has been sitting 5 quarters - past the 4-quarter mark where we start treating a deal as stale. Its chance of closing is 0.0% rather than 0.0% for a fresh deal.

• Your file holds 376,539 rows. We are running on an evenly spread sample of 188,272 of them (50.0%), keeping rows intact rather than splitting a record down the middle, because a working set that big does not fit in memory while keeping the tool responsive. Every number below comes from those 188,272 rows.

Assumptions

• Learned how deals hop between stages from 100,958 real stage moves (the quarter column told us what happened first). Paths nobody has taken yet are given a small benefit of the doubt rather than a flat zero chance.

• A deal that is won or dead stays that way - we force those stages to be final even if the history is messy.

• We priced how long deals sit still: past 4 quarters a deal gets progressively more likely to slip or die, up to a cap of 90.0% extra risk. The deal being forecast is treated as 5 quarters old.

• Deal value $3,000,000, starting in 'Initial RFP', 6 quarters ahead, fiscal year = 4 quarters.

• Each quarter a deal moves at most one stage forward; it may sit still or die. Deal size does not change the odds.

• Stage behaviour is assumed stable: the pattern in your history continues.

• Columns of your file we actually used: deal_id, stage, stage_order, quarter, amount, entered_quarter, days_in_stage.

• Deal / opportunity ID -> your 'deal_id' column

• Pipeline stage -> your 'stage' column

• Stage order (1,2,3...) -> your 'stage_order' column

• Quarter of this sighting -> your 'quarter' column

• Deal value -> your 'amount' column

• Quarter the deal entered -> your 'entered_quarter' column

• How long the deal has been sitting (quarters) -> not in your file; not used

• How long it has been sitting (days) -> your 'days_in_previous_stage' column

• Data used: 188,272 of the 376,539 rows in your file (50.0%), drawn evenly across it.

Appendix: the last steps of the run

Every line was produced while the model ran. Re-running with the same seed replays it exactly.

Time	Stage	What happened
0.01s	prepare	You chose 'deal_id' for 'Deal / opportunity ID'.
0.01s	prepare	You chose 'stage' for 'Pipeline stage'.
0.01s	prepare	You chose 'stage_order' for 'Stage order (1,2,3...)'.
0.01s	prepare	You chose 'quarter' for 'Quarter of this sighting'.
0.01s	prepare	You chose 'amount' for 'Deal value'.
0.01s	prepare	You chose 'entered_quarter' for 'Quarter the deal entered'.
0.01s	prepare	We could not find a 'How long the deal has been sitting (quarters)' column. We left it out and will work without it.
0.01s	prepare	You chose 'days_in_previous_stage' for 'How long it has been sitting (days)'.
0.01s	prepare	Deal / opportunity ID -> your 'deal_id' column
0.01s	prepare	Pipeline stage -> your 'stage' column
0.01s	prepare	Stage order (1,2,3...) -> your 'stage_order' column
0.01s	prepare	Quarter of this sighting -> your 'quarter' column
0.01s	prepare	Deal value -> your 'amount' column
0.01s	prepare	Quarter the deal entered -> your 'entered_quarter' column
0.01s	prepare	How long the deal has been sitting (quarters) -> not in your file; not used
0.01s	prepare	How long it has been sitting (days) -> your 'days_in_previous_stage' column

Time	Stage	What happened
3.44s	prepare	Read 188,272 records into the model.
3.44s	simulate	Running the simulation…
3.44s	read	Reading how your deals have moved through the pipeline...
5.42s	prepare	Laid the stages out in the order you gave us.
5.42s	prepare	Learned the hop pattern from 100,958 real stage moves across 87,314 deals.
5.42s	prepare	Deals that sit still get riskier: this one has been going 5 quarters, so we priced it against the stale-deal pattern.
5.42s	simulate	Rolling the odds forward 6 quarters from 'Initial RFP'...
5.42s	summarize	Sales says quarter 2. The odds from your history are 0.0% for closing in that quarter.
6.49s	summarize	Done. Plan on $32 of $3.00M for Q2, not the full amount.